

Nationality
French
Current home
Oxford, UK
Website
Personal 

Edith Darin

Demography, Bayesian statistics and Geospatial data science

Contact

E-mail
edith.darin@
protonmail.com

Tech repo

Github
edarin
OpenStreetMap
edithdarin

Programming

OS Preferences
Windows
GNU/Linux
Statistical Programming
R (+++)
Python (++)
Dashboard & API
Docker, Flask,
PostgreSQL, DuckDB,
R Shiny, Git, Nginx
GIS & Remote Sensing
QGIS, Google Earth
Engine, GDAL,
PostGIS

Languages

French native
English fluent
German intermediate
Italian intermediate

Work

- 2026 - **Population Data Scientist** Demographic Unit, University of Oxford, UK
2024 -
- *Secratio project*: technical lead for data infrastructures linking survey data to daily environmental stressors (temperature, air pollution, conflict events). *Methods developed*: Duckdb interface for Parquet files. PI: Dr Joshua Wilde
- *NowPop project*: Model monthly forced internal displacement in war-struck Ukraine. *Methods developed*: Statistic modelling of flows of mobile phone subscribers. Automated dockerised data pipeline. Deploy nginx dashboard and flask API for dissemination. Partner: WHO. PI: Dr Douglas Leasure
- 2024 - **Research Fellow** WorldPop, University of Southampton, UK
2018 -
Until 08/21 *Specialist Research Technician*
Lead modeller for filling population gaps in censuses impacted by insecurity (Burkina Faso, Mali, Democratic Republic of Congo). Map refugee population in Cameroon. Partners: UNFPA, UNHCR, Gates foundation, CIESIN.
Methods developed: Hierarchical Bayesian modelling combining survey data with geospatial covariates. Random-forest models for producing high-resolution population estimates. Grid-based spatial sampling design for household surveys.
- 2017 **Data Scientist Intern** mVAM, UN World Food Program, Italy
Assessing food security from remote sensing and mobile data collection. Machine-learning based propensity score weighting in complex survey design. Support mission for Mali Country Office (Bamako).
- 2016 **Data Scientist Intern** Etalab, Governmental Task Force for OpenData, France
OpenFisca: French tax and benefit system micro-simulation platform. Finalist of the Open Government Partnership Hackathon.

Grants

- 2021 **Mapping refugees with high spatial resolution** UNHCR: \$80 000
Co-I for developing methods to map refugees for survey purposes.
- 2022 **Statistical population modelling for census support** UNFPA: \$30 000
Co-I for creating a population model with Brazil Statistics Office (IBGE).

Education

- 2026 - **DPhil in Demography** University of Oxford, UK
2022 -
Topic: *Estimating subnational population sizes in data-scarce contexts*. Affiliated with the Max Planck Institute for Demographic Research and in collaboration with the Colombian Stats Office
- 2019 - **MSc in Applied GIS and Remote Sensing** University of Southampton, UK
2018 -
Dissertation featured in a peer-reviewed article.
- 2018 - **MSc in Statistics and Economics** ENSAE, France
2015 -
Semesters abroad at Humboldt University and Galatasaray University.

Academic outcomes

Peer-reviewed papers

- Leveraging high-frequency digital data to analyze forced displacement dynamics: A case study from the Gaza Strip. **Darin E**, Kashyap, R. and Leasure, D.R. *Population and Development Review* 12(4), 2026. [↗](#)
- Developing a customized, enumeration area-based sampling frame tailored to a specific population subgroup using geospatial methods, Qader S., **Darin E.**, Dicko A.H., Galal H., Park H., Jimenez R.M., Harfoot A., and Tatem A.J. *Journal of Survey Statistics and Methodology* 14(1), 2026. [↗](#)
- Demographic figures at risk in the digital era: Resisting commodification, reclaiming the common good. **Darin E.** *Big Data & Society* 12(4), 2025. [↗](#)
- How accurate are high-resolution settlement maps at predicting population counts in data-scarce settings?, **Darin E.**, Kashyap, R. and Leasure, D.R. *Population, Space and Place*, 31: e70083, 2025. [↗](#)
- Tackling public health data gaps through Bayesian high-resolution population estimation: A case study of Kasai-Oriental, Democratic Republic of the Congo, Boo G, **Darin E**, Chamberlain HR, Hosner R, Akilimali PK, et al. 2025 *PLOS Global Public Health*, 5(9): e0005072. [↗](#)
- Mapping refugee populations at high-resolution by unlocking humanitarian administrative data, **Darin, E.**; Dicko D. H.; Galal H.; Jimenez R.M.; Park H.; Tatem A.J.; Qader S., *Journal of International Humanitarian Action*, 2024 [↗](#)
- Building footprint data for countries in Africa: To what extent are existing data products comparable?, Chamberlain H. R.; **Darin, E.**; Adewole A. W.; Lazar A.; Tatem A.J., *Computers, Environment and Urban Systems*, 2024, 102104, [↗](#)
- The Population Seen from Space: When Satellite Images Come to the Rescue of the Census, **Darin, E.**; Kuépié M.; Bassinga H, Boo G., Tatem A.J.; *Population*, 2022, 77:3, [↗](#)
- High-resolution population estimation using household survey data and building footprints, Boo, G; **Darin, E.**; Leasure, D.R.; Dooley, C.A.; Chamberlain, H.R., et al, *Nature Communications*, 2022, 13:1, [↗](#)
- The role of earth observation in an integrated deprived area mapping "system" for low-to-middle income countries, Kuffer, M.; Thomson, D.R.; Boo, G.; Mahabir, R.; Grippa, T.; Vanhuysse, S.; Engstrom, R.; Ndugwa, R.; Makau, J.; **Darin, E.**; de Albuquerque, J.P. and Kabaria, C, *Remote Sensing*, 2020, 12, 982. [↗](#)
- A grid-based sample design framework for household surveys, Boo G; **Darin E**; Thomson DR and Tatem AJ, *Gates Open Research* 2020, 4:13, [↗](#)

Pre-prints

- Lazar, A. N.; Boo, G.; Chamberlain, H. R.; Nnanatu, C. C.; **Darin, E.**; Leasure, D. R.; Yankey, O.; Gadiaga, A.; Juran, S.; de la Rua, L.; Espey, J.; Tatem, A. J. Advances in Small Area Population Estimation in the Absence of National Census Data. Preprints 2025, 2025061625. [↗](#)

Book chapter

- Demographic figures at risk in the digital era: Resisting commodification, reclaiming the common good **Darin, E.** in *Digital Fairness and data protection in the age of artificial intelligence*, Lefebvre Giuffre. [↗](#)

R Package

- wopr: An R package to query the WorldPop Open Population Repository, version 1.0.1, Leasure DR; Bondarenko M; **Darin E**; Tatem AJ, *WorldPop, University of Southampton*, 2020. Maintainer: **Darin E.** [↗](#)

Reviews

I have reviewed papers for *Data and Policy*, *Geoenvironmental disasters*, *Habitat international*, *PLOS Global Health*, *Royal Society OpenScience*, and *Vienna Yearbook for Population Research*.

Committee position

I am a member of the Royal Statistical Society Data Ethics and Governance Section Committee

Recent conferences

- Oral: Statistical Innovation for Population Estimation: Integrating Administrative Records, Geospatial Data, and Real-Time Streams, **Darin E**, *EEAMO Conference*, 2025.
- Poster: Estimating intercensal populations with incomplete administrative data: a state-space Leslie matrix approach, **Darin E**, *EvoDemo*, 2025.
- Oral: Unraveling the Impact of the 2023 Conflict between Israel and Hamas on Short And Mid-Term Population Internal Displacements in the Middle East, **Darin E**, *European Population Conference*, 2024.
- Oral: Assessing Satellite Imagery Performance for Population Estimation in Data-Scarce Settings: A Modeling Perspective, **Darin E**, *Population Association of America*, 2024.
- Oral: Estimating population in data-scarce contexts: to which extent can we leverage satellite imagery, **Darin E**, *British Society of Population Studies*, 2023.
- Oral: Counting people from space: an inquiry from a data practitioner perspective, **Darin E**, *Data Justice Conference*, 2023.

Teaching

- Workshop: Gridded population modelling for Summer Institutes in Computational Social Science, Covenant, Nigeria 2022, Joseph Ki-Zerbo and Norbert Zongo University, Burkina Faso 2021, Data Science Nigeria Bootcamp 2019.
- Tutorials: Statistical population modelling for census support; *WorldPop*, *University of Southampton*, 2021, [↗](#). Small area population estimates using random forest top-down disaggregation: An R tutorial; *WorldPop*, *University of Southampton*, 2020. [↗](#).
- R Foundations Course for different Nigerian national agencies, 2020. Population Modelling Webinar, *GRID3 Webinar Series*, 2020. [↗](#)

Awards

- Julia Mead Knox Memorial Prize for Research in Demography, University of Oxford (1000£), 2025
- Three-year scholarship from the Leverhulme Centre for Demographic Research (3x32,000), 2022, Light Senior scholarship, St Catherine's college, 2024
- Vice-Chancellor Award for Research Impact (2018), Faculty Dean's Prize for Research (2021), University of Southampton
- Four-year scholarship from the École Normale supérieure de Cachan (4x15,600€), 2015, Scholarship Sarah Andrieux for Social and Economics Research (800€), 2017